

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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Chemistry

- 1 Three liquid fractions (P, Q, and R) are extracted from a biological sample. Their physical properties are shown in the table below.

Fraction	Density / g cm ⁻³	Boiling point / °C	Miscibility
P	0.79	56	Miscible with Q
Q	1.00	100	Miscible with P
R	1.49	61	Immiscible with P and Q

The fractions are mixed in a separating funnel and allowed to settle, forming two layers.

Which procedure describes the correct method to obtain a pure sample of fraction P?

- A Collect the upper layer and heat to 56°C
 - B Collect the upper layer and heat to 100°C
 - C Collect the lower layer and heat to 56°C
 - D Collect the lower layer and heat to 61°C
 - E Collect the upper layer and heat to 61°C
- 2 A student conducts two experiments to heat liquids using the energy released from the combustion of a fixed amount of ethanol. Assume that there is no heat loss and that the amount of energy transferred to the liquid is the same in both experiments.

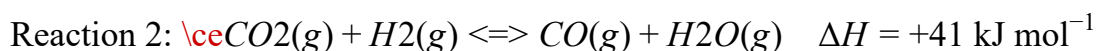
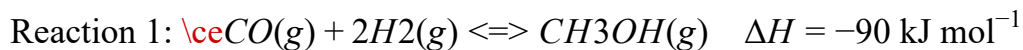
In Experiment 1, mass m of liquid X (specific heat capacity c) is heated, and the temperature rises by ΔT .

In Experiment 2, mass $3m$ of liquid Y (specific heat capacity $\frac{1}{2}c$) is heated.

Which expression gives the temperature rise in Experiment 2?

- A $\frac{1}{6}\Delta T$
- B $\frac{1}{3}\Delta T$
- C $\frac{2}{3}\Delta T$
- D $\frac{3}{2}\Delta T$
- E $6\Delta T$

- 3 Two reversible gas-phase reactions are carried out in separate closed vessels:



The following actions could be applied independently to each reaction system:

- 1 decreasing the volume of the reaction container at constant temperature
- 2 increasing the temperature of the reaction container at constant volume
- 3 adding an appropriate catalyst at constant temperature and volume

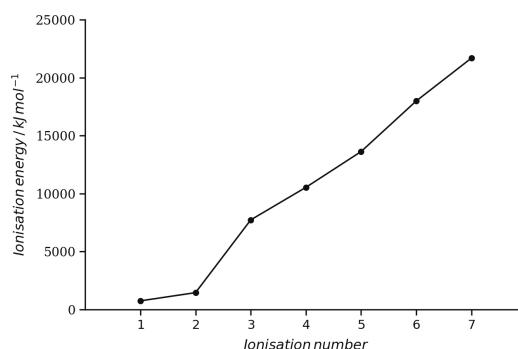
Assuming all gases behave ideally, which of these actions will increase the initial rate of the forward reaction for **both** reactions, **and** increase the equilibrium yield of products for Reaction 1, while having **no effect** on the equilibrium yield of products for Reaction 2?

- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

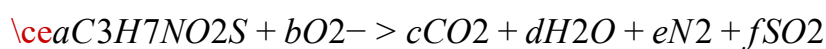
- 4 The graph shows the first seven successive ionisation energies of an element X , which belongs to Period 3 of the Periodic Table.

Which row in the table correctly identifies the formula of the simple oxide of X , the structure and bonding present in this oxide, and its acid-base character?

	formula of oxide	structure and bonding	acid-base character
A	X_2O	giant ionic	basic
B	X_2O	simple covalent	acidic
C	XO	giant ionic	basic
D	XO	simple covalent	acidic
E	XO_2	giant covalent	acidic
F	XO_2	simple covalent	acidic
G	X_2O_3	giant ionic	basic
H	X_2O_3	giant ionic	amphoteric



- 5 The complete combustion of the amino acid cysteine, $C_3H_7NO_2S$, produces carbon dioxide, water, nitrogen gas, and sulfur dioxide according to the equation:

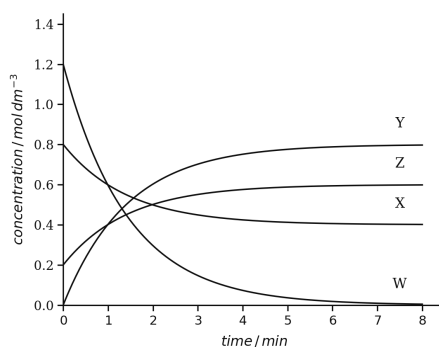


What is the value of b when the equation is balanced using the lowest whole-number coefficients?

- A 9
- B 11
- C 15
- D 17
- E 19
- F 23
- G 38

- 6 The graph shows how the concentrations of four substances, W , X , Y , and Z , change over time during a reaction in a closed vessel at constant temperature.

Which of the following is the balanced equation for this reaction?

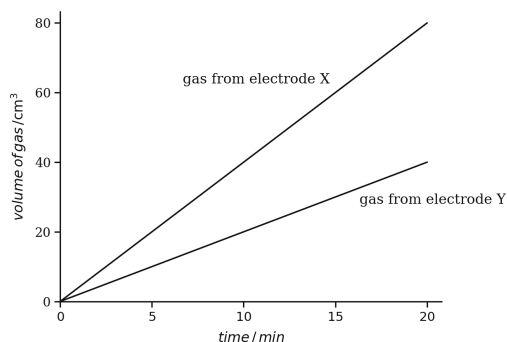


- A $3W + X \rightarrow 2Y + Z$
 B $3W + X \rightarrow 4Y + 3Z$
 C $3W + 2X \rightarrow 2Y + Z$
 D $3W + 2X \rightarrow 2Y + 3Z$
 E $6W + 4X \rightarrow 4Y + 3Z$
 F $3W \rightarrow X + 2Y + Z$

- 7 The graph shows the volume of gas collected at each of two inert electrodes, X and Y, during the electrolysis of dilute aqueous sulfuric acid, $\text{H}_2\text{SO}_4(\text{aq})$, at constant current, temperature, and pressure.

Which of the following statements is/are correct?

1. Electrode X is connected to the negative terminal of the power supply.
2. In the external circuit, electrons flow from electrode X to electrode Y.
3. For every 1 mol of gas collected at electrode Y, 4 mol of electrons pass through the external circuit.



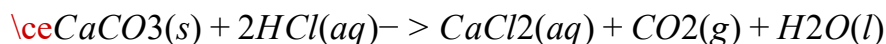
- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 8 The following information about four non-metallic elements labelled W, X, Y and Z is given:
- Elements W and Y displace bromine from an aqueous solution of potassium bromide, KBr(aq) .
 - Elements X and Z cannot displace bromine from an aqueous solution of potassium bromide, KBr(aq) .
 - Element Y reacts explosively with hydrogen gas in the dark, whereas element W requires ultraviolet light to react with hydrogen gas.
 - Element Z reacts with an aqueous solution of the sodium salt NaX to displace element X.

What is the order of reactivity (oxidising ability) of these four elements, starting with the most reactive?

- A W, Y, X, Z
 - B W, Y, Z, X
 - C X, Z, W, Y
 - D X, Z, Y, W
 - E Y, W, X, Z
 - F Y, W, Z, X
 - G Z, X, W, Y
 - H Z, X, Y, W
- 9 Element X has an atomic number of 15. It reacts with hydrogen to form a simple molecular hydride.
- Which option gives the correct formula of the hydride and the shape of the molecule?
- A XH_2 , linear
 - B XH_2 , bent (V-shaped)
 - C XH_3 , trigonal planar
 - D XH_3 , trigonal pyramidal
 - E XH_4 , tetrahedral

- 10 Calcium carbonate reacts with hydrochloric acid according to the equation:



In a baseline experiment (represented by curve **X** on the graph), 2.00 g of large marble chips (CaCO_3) reacts with 50 cm³ of 1.0 mol dm⁻³ $\text{HCl}(aq)$ at 25 °C.

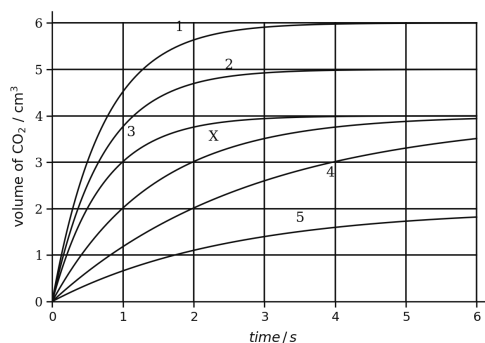
Two further experiments are carried out at 25 °C:

- **Experiment 1:** 3.00 g of powdered CaCO_3 is reacted with 50 cm³ of 1.0 mol dm⁻³ $\text{HCl}(aq)$
- **Experiment 2:** 2.00 g of large marble chips (CaCO_3) is reacted with 100 cm³ of 0.40 mol dm⁻³ $\text{HCl}(aq)$

(M_r value: $\text{CaCO}_3 = 100$)

Which curves show how the volume of CO_2 evolved changes with time in Experiment 1 and Experiment 2?

	Experiment 1	Experiment 2
A	curve 1	curve 4
B	curve 1	curve 5
C	curve 2	curve 4
D	curve 2	curve 5
E	curve 3	curve 4
F	curve 3	curve 5

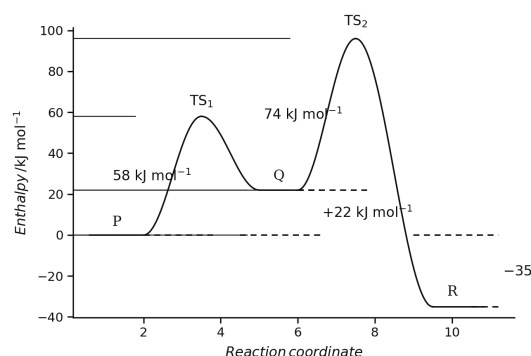


- 11 The reaction energy profile for a two-step gas-phase reaction $P(g) \rightarrow Q(g) \rightarrow R(g)$ is shown in the diagram below.

From the information provided in the energy profile:

- The activation energy for the forward step $P \rightarrow Q$ is 58 kJ mol^{-1} .
- The intermediate Q has an enthalpy 22 kJ mol^{-1} higher than reactant P .
- The activation energy for the forward step $Q \rightarrow R$ is 74 kJ mol^{-1} .
- The overall enthalpy change for $P \rightarrow R$ is $\Delta H = -35 \text{ kJ mol}^{-1}$.

What are the activation energy for the conversion of intermediate Q to reactant P ($E_a(Q \rightarrow P)$) and the activation energy for the conversion of product R to intermediate Q ($E_a(R \rightarrow Q)$)?



- A $E_a(Q \rightarrow P) = 36 \text{ kJ mol}^{-1}$, $E_a(R \rightarrow Q) = 61 \text{ kJ mol}^{-1}$
 B $E_a(Q \rightarrow P) = 36 \text{ kJ mol}^{-1}$, $E_a(R \rightarrow Q) = 109 \text{ kJ mol}^{-1}$
 C $E_a(Q \rightarrow P) = 36 \text{ kJ mol}^{-1}$, $E_a(R \rightarrow Q) = 131 \text{ kJ mol}^{-1}$
 D $E_a(Q \rightarrow P) = 58 \text{ kJ mol}^{-1}$, $E_a(R \rightarrow Q) = 109 \text{ kJ mol}^{-1}$
 E $E_a(Q \rightarrow P) = 58 \text{ kJ mol}^{-1}$, $E_a(R \rightarrow Q) = 131 \text{ kJ mol}^{-1}$
 F $E_a(Q \rightarrow P) = 80 \text{ kJ mol}^{-1}$, $E_a(R \rightarrow Q) = 39 \text{ kJ mol}^{-1}$
 G $E_a(Q \rightarrow P) = 80 \text{ kJ mol}^{-1}$, $E_a(R \rightarrow Q) = 109 \text{ kJ mol}^{-1}$
 H $E_a(Q \rightarrow P) = 80 \text{ kJ mol}^{-1}$, $E_a(R \rightarrow Q) = 131 \text{ kJ mol}^{-1}$

- 12 A mixture containing three volatile organic compounds, P , Q , and R , is analysed using gas chromatography. The resulting chromatogram is shown below.

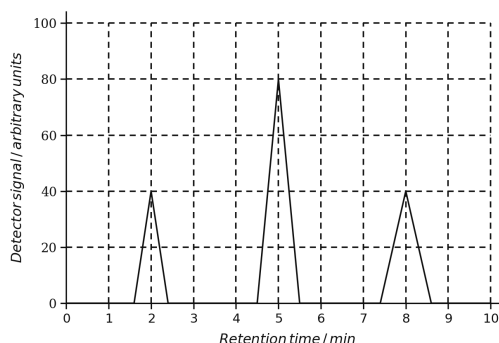
Standard samples of the pure compounds analysed under identical conditions gave the following retention times:

- Compound P : 2.0 min
- Compound Q : 5.0 min
- Compound R : 8.0 min

Assume that the detector response per unit mass is identical for all three compounds, and that each peak can be approximated as a triangle (area = $\frac{1}{2} \times \text{base width} \times \text{height}$).

Which of the following statements is/are correct?

- 1 Compound Q makes up 50% of the total mass of the mixture.
- 2 Under these conditions, compound R has a stronger affinity for the stationary phase than compound P .
- 3 Compound P makes up 25% of the total mass of the mixture.



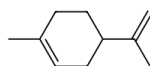
- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

13 The skeletal structure of limonene is shown below:

Limonene undergoes complete catalytic hydrogenation to form a fully saturated hydrocarbon.

What is the minimum volume of hydrogen gas, measured at room temperature and pressure (RTP), required to react completely with 1.60 cm^3 of limonene?

(M_r of limonene = 136; density of limonene = 0.85 g cm^{-3} ; molar volume of a gas at RTP = $24.0 \text{ dm}^3 \text{ mol}^{-1}$)



- A** 0.024 dm^3
- B** 0.048 dm^3
- C** 0.24 dm^3
- D** 0.48 dm^3
- E** 0.72 dm^3
- F** 0.96 dm^3
- G** 2.4 dm^3
- H** 4.8 dm^3

- 14** A 0.45 g sample of the organic compound shown below reacts completely with an excess of sodium metal at room temperature.

Any gas produced is collected and its volume is measured at room temperature and pressure (r.t.p.).

Assuming 1.0 mol of gas occupies 24.0 dm^3 at r.t.p., what volume of gas is collected?

(A_r values: $H = 1.0$; $C = 12.0$; $O = 16.0$)



- A** 0.12 cm^3
- B** 60 cm^3
- C** 120 cm^3
- D** 180 cm^3
- E** 240 cm^3
- F** 360 cm^3
- G** 600 cm^3

- 15** Isolated mitochondria are suspended in a buffer containing pyruvate as the sole respiratory substrate. The table below shows the net yield of reduced coenzymes and $ceCO_2$ for the relevant stages of respiration.

Stage	Reduced NAD	Reduced FAD	$ceCO_2$
Link reaction (per pyruvate)	1	0	1
Krebs cycle (per acetyl CoA)	3	1	2

In the electron transport chain, the oxidation of one molecule of reduced NAD or reduced FAD releases two electrons. Four electrons are required to reduce one molecule of oxygen (ceO_2).

If the mitochondrial suspension produces 3.0 moles of $ceCO_2$, how many moles of ceO_2 are consumed?

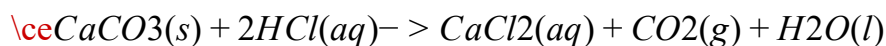
- A** 2.0
- B** 2.5
- C** 3.0
- D** 5.0

- 16** Propan-2-ol is warmed with acidified potassium dichromate(VI) solution. Which of the following statements about this reaction is/are correct?

- 1 The reaction is classified as a substitution reaction.
- 2 The organic product gives a positive test with Tollens' reagent.
- 3 The organic product is a ketone.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 17 An experiment investigates the reaction between solid marble chips (calcium carbonate, CaCO_3) and an excess of dilute hydrochloric acid:



In a second experiment, the same mass of CaCO_3 is used, but the solid is ground into a fine powder. All other conditions, including the volume, concentration, and initial temperature of the hydrochloric acid, are kept exactly the same.

How do the frequency of collisions between reactant particles, the activation energy of the reaction, and the initial rate of reaction in the second experiment compare to those in the first experiment?

	Frequency of collisions between reactant particles	Activation energy	Initial rate of reaction
A	decreases	decreases	decreases
B	decreases	unchanged	decreases
C	unchanged	decreases	increases
D	unchanged	unchanged	increases
E	increases	decreases	increases
F	increases	increases	increases
G	increases	unchanged	increases
H	increases	unchanged	unchanged

- 18 Concentrated aqueous solutions of the following three compounds are electrolysed separately using inert electrodes: 1.

NaCl 2.

HBr 3.

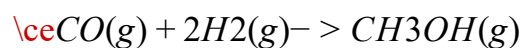
CuCl_2

Which of the solutions yield(s) the elemental form of **all** the constituent elements of the solute at the electrodes?

- A 3 only
- B 1 and 2 only
- C 1 and 3 only
- D 2 and 3 only
- E 1, 2 and 3

- 19** A redox reaction occurs in acidic solution between the complex ion PtCl_4^{2-} and the permanganate ion MnO_4^- . In this reaction:
- * The platinum is oxidised from its state in the complex ion to the +4 oxidation state. * The manganese is reduced from its state in the permanganate ion to the +2 oxidation state.
- What is the simplest whole number mole ratio of PtCl_4^{2-} to MnO_4^- required for the reaction?
- A** 1 : 1
B 2 : 1
C 2 : 5
D 5 : 2
E 5 : 1
- 20** A sample of a metal M consists of only two isotopes: ^{69}M (relative isotopic mass 69.0) and ^{71}M (relative isotopic mass 71.0).
- A 4.872 g sample of pure metal M reacts completely with oxygen to form 6.552 g of the oxide M_2O_3 .
- What is the percentage abundance of the ^{69}M isotope in this sample?
- (Assume the relative atomic mass of oxygen is 16.0.)
- A** 30%
B 35%
C 60%
D 65%
E 70%
F 75%

- 21 Methanol can be synthesised industrially in the gas phase from carbon monoxide and hydrogen according to the following equation:



When 1 mol of CO(g) reacts completely with hydrogen, 112 kJ of energy is released.

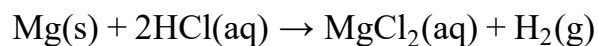
Some mean bond enthalpies are given in the table below:

Bond	Mean bond enthalpy / kJ mol ⁻¹
C–H	413
C–O	358
O–H	464
H–H	436

What is the bond enthalpy of the carbon–oxygen bond in the carbon monoxide molecule?

- A 613 kJ mol⁻¹
- B 965 kJ mol⁻¹
- C 1077 kJ mol⁻¹
- D 1189 kJ mol⁻¹
- E 1301 kJ mol⁻¹
- F 1513 kJ mol⁻¹

- 22** Magnesium reacts with hydrochloric acid according to the equation:



Five experiments are set up to produce hydrogen gas.

Exp	Mass Mg / g	Vol HCl / cm ³	Conc HCl / mol dm ⁻³	State of Mg	Temp / °C
1	1.2	200	2.0	Powder	60
2	3.6	150	1.0	Powder	50
3	2.4	300	1.0	Ribbon	20
4	2.4	100	2.0	Powder	40
5	4.8	100	2.0	Ribbon	40

(Ar value: Mg = 24)

Which experiment produces at least 0.10 mol of H₂ in the shortest time?

- A** Experiment 1
- B** Experiment 2
- C** Experiment 3
- D** Experiment 4
- E** Experiment 5

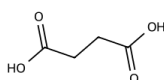
23 An organic carboxylic acid **X** has the structural formula shown below:

A student dissolves 0.590 g of acid **X** in distilled water and makes the volume up to 250 cm³ in a volumetric flask.

A 25.0 cm³ portion of this solution is transferred to a conical flask and titrated against aqueous potassium hydroxide, KOH(aq) . Exactly 20.0 cm³ of the KOH(aq) solution is required for complete neutralisation.

What is the concentration of the potassium hydroxide solution in mol dm⁻³?

(A_r values: $H = 1.0$, $C = 12.0$, $O = 16.0$)



- A** 0.0125 mol dm⁻³
- B** 0.0250 mol dm⁻³
- C** 0.0500 mol dm⁻³
- D** 0.100 mol dm⁻³
- E** 0.250 mol dm⁻³
- F** 0.500 mol dm⁻³

- 24 An addition polymer is formed from a hydrocarbon monomer C_6H_{12} with the molecular formula C_6H_{12} .

In the repeating unit of this polymer:

- the main chain (polymer backbone) consists of 2 carbon atoms
- every side group attached to the main chain is a methyl group (CH_3)
- neither of the two carbon atoms in the main chain is bonded to more than one hydrogen atom.

Which of the following is the systematic IUPAC name for C_6H_{12} ?

- A hex-1-ene
 B hex-2-ene
 C hex-3-ene
 D 2-methylpent-2-ene
 E 3-methylpent-2-ene
 F 2,3-dimethylbut-1-ene
 G 2,3-dimethylbut-2-ene
 H 3,3-dimethylbut-1-ene
- 25 Element X is in Period 3 of the Periodic Table. In its ground state, an atom of X has three unpaired electrons, and it reacts with chlorine to form a chloride with the formula XCl_5 .

Which row of the table correctly describes the elements adjacent to X in the Periodic Table?

	element immediately above X	element immediately below X	element immediately to the left of X	element immediately to the right of X
A	exists as diatomic molecules containing a triple bond	has a larger atomic radius than X	forms an oxide with a giant covalent structure	has a higher first ionisation energy than X
B	can expand its octet to form a pentachloride	has a larger atomic radius than X	contains one more proton in its nucleus than X	has a lower first ionisation energy than X
C	exists as diatomic molecules containing a triple bond	has a larger atomic radius than X	forms an oxide with a giant covalent structure	has a lower first ionisation energy than X
D	has a lower first ionisation energy than X	contains fewer occupied electron shells than X	forms an oxide with a giant covalent structure	has a higher first ionisation energy than X
E	exists as diatomic molecules containing a triple bond	has a higher first ionisation energy than X	contains one more proton in its nucleus than X	has a lower first ionisation energy than X

- 26** An ion with the formula $\text{Q}_4\text{O}_6^{2-}$ consists of a linear chain of four atoms of element Q. The two central Q atoms are bonded only to other Q atoms. The two terminal Q atoms are each bonded to one Q atom and three oxygen atoms.

Which option gives the oxidation states of the Q atoms in this ion?

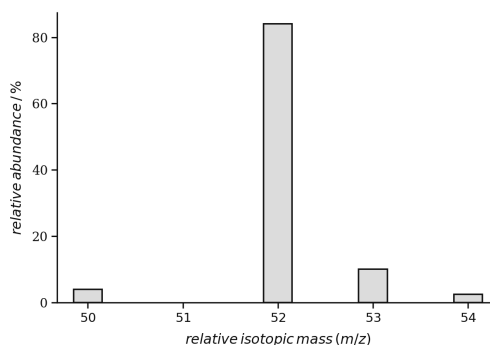
- A** +2.5 only
- B** 0 and +5
- C** 0 and +6
- D** +2 and +3
- E** +4 and +6

- 27** The mass spectrum of an element X is shown in the graph below.

A 0.100 mol sample of element X reacts completely with 0.075 mol of $\text{O}_2(\text{g})$ to form a single oxide.

Naturally occurring oxygen consists of three isotopes with mass numbers 16, 17, and 18.

Using this information, what is the difference between the maximum possible and the minimum possible relative formula mass of this oxide?



- A** 6
- B** 8
- C** 10
- D** 12
- E** 14
- F** 16
- G** 22